Chapter

8 Acids and Bases

1. **[7 marks]** *2.8 (2015:29)*

A 25.0 mL solution of nitric acid at 25.0 °C contains 8.50×10^{-3} moles of hydrogen ions.

(a) Calculate the hydrogen ion concentration and the pH of the solution. [2]

(b) Calculate the pH of the solution after 20.0 mL of 0.300 mol L^{-1} potassium hydroxide solution is added to the original 25.0 mL of nitric acid. [5]

2. **[15 marks]** *(2015:40)*

Hydrogen fluoride, HF, is a highly dangerous and corrosive liquid that boils at near room temperature. It readily forms hydrofluoric acid in the presence of water and is an ingredient used to produce many important compounds, including medicines and polymers.

(a) (i) Name the electrostatic attractive force that holds the hydrogen and fluorine atoms together **within** hydrogen fluoride molecules. [1]

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(ii) Name the electrostatic attractive force **between** the hydrogen fluoride molecules. [1]

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(iii) Explain the origin of the attractive force **between** the hydrogen fluoride molecules. [2]

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(b) The equilibrium constant (K) for the dissociation of hydrofluoric acid is 6.8×10^{-4}, and for hydrochloric acid K is very large. To make a solution of hydrofluoric acid with the same pH as hydrochloric acid, a greater concentration of hydrofluoric acid is required. Explain why this is so. [3]

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(c) The salts, sodium chloride and sodium fluoride, readily dissolve in water. At 25.0 °C the pH of the sodium chloride solution is equal to 7 whereas the pH of the sodium fluoride solution is greater than 7. Explain this difference in pH. Include any relevant equation(s) to support your answer. [3]

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Propanoic acid, CH_3CH_2COOH, is also a weak monoprotic acid. When 0.500 mol of sodium propanoate is dissolved in 1.00 L of 0.500 mol L^{-1} propanoic acid at 25.0 °C a buffer solution is formed.

(d) (i) Addition of 10.0 mL of 1.00 mol L^{-1} $HC\ell(aq)$ to this buffer does not significantly change its pH. Explain this observation, including any relevant equation(s). [3]

(ii) State **two** conditions required to ensure that this system has a high buffering capacity. [2]

One:

Two:

3. **[6 marks]** *2.4 (2016:31)*

(a) Select **one** basic, **one** acidic and **one** neutral salt from the list below to complete the table. [3]

KCN, $NH_4C\ell$, $Mg_3(PO_4)_2$, $NaNO_3$, $KHCO_3$, $NaCH_3COO$, $KC\ell$

Acidic salt	Neutral salt	Basic salt

(b) Use the Brønsted-Lowry model to explain why the pH of ammonia solution is greater than 7.0 at 25 °C. Incorporate at least **one** appropriate equation in your answer. [3]

4. **[6 marks]** *2.5 (2016:32)*

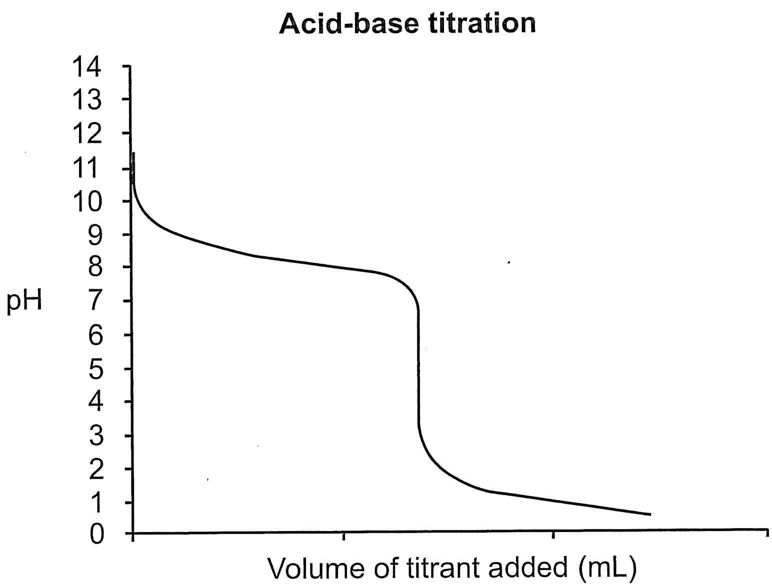
(a) A buffer of carbonic acid (H_2CO_3)/hydrogencarbonate (HCO_3^-) is present in blood plasma to maintain a pH between 7.35 and 7.45. Write an equation to show the relevant species present in a carbonic acid/hydrogencarbonate buffer solution. [2]

(b) Explain why 300.0 mL of 1.00 mol L^{-1} carbonic acid/hydrogencarbonate buffer does **not** change in pH significantly when 3 drops of 1.00 mol L^{-1} $HC\ell$ are added to it, yet when 3 drops of 1.00 mol L^{-1} $HC\ell$ are added to 300.0 mL of distilled water there is a significant change in pH. [4]

5. **[6 marks]** *2.01 (2016:34)*

The data below were collected from an acid-base titration.

(a) Label the equivalence point on the titration curve below using an arrow and record the pH value at this point. [2]



pH value at equivalence point: ______________________

(b) Select an indicator from the table below that would be **best** for this titration and justify your choice. [4]

Indicator	Low pH colour	Transition pH range	High pH colour
Methyl Yellow	red	2.1 – 3.3	yellow
Bromocresol Green	yellow	3.8 – 5.4	blue
Bromothymol Blue	yellow	6.0 – 7.6	blue
Phenolphthalein	colourless	8.3 – 10.0	pink
Alizarin Yellow R	yellow	10.2 – 12.0	red

Indicator: ______________________

Justification: ______________________

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6. **[8 marks]** *2.5 (2017:31)*

Water is capable of self-ionisation.

(a) Write an equation for the self-ionisation of water. [2]

(b) Write the equilibrium constant expression for the self-ionisation of water. [1]

(c) The equilibrium constant for the self-ionisation of water, Kw, is 1.00×10^{-14} at 25 °C. What does this value indicate about this reaction? [1]

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The K values for the self-ionisation of water at 100.0 kPa are given here for a number of different temperatures.

Temperature (°C)	**K value**
0	0.114×10^{-14}
25	1.00×10^{-14}
50	5.48×10^{-14}
75	19.9×10^{-14}
100	51.3×10^{-14}

(d) Calculate the pH of water at 50 °C. [2]

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(e) Is water acidic, basic or neutral at 50 °C? State a reason for your answer. [2]

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7. **[10 marks]** *(2018:26)*

Solid copper(II) hydroxide is added to excess 0.100 mol L^{-1} carbonic acid solution.

(a) Write the balanced equation, with appropriate state symbols, for the reaction that takes place between the copper(II) hydroxide and carbonic acid. [3]

(b) Predict **all** visible changes that would be observed, if any, while the reactants are mixed together and afterwards. [3]

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(c) Predict **two** observations that would be different if excess 0.100 mol L^{-1} hydrochloric acid was used instead of the 0.100 mol L^{-1} carbonic acid. [2]

One: ____

Two: ____

(d) State **two** personal safety measures the experimenter should take when conducting these experiments. [2]

One: ____

Two: ____

8. **[12 marks]** *(2018:27)*

Phosphoric acid, $H_3PO_4(aq)$, is a weak, triprotic acid.

(a) Write the ionisation equation for phosphoric acid in water which shows the **second** proton of the acid being released into solution. [2]

Magnesium carbonate, $MgCO_3(s)$, is an ingredient of a commonly-used antacid.

(b) Other than water, list **three** species (elements, compounds, ions) that would be found in the reacting vessel open to the atmosphere at the completion of the reaction between excess solid magnesium carbonate and an aqueous solution of phosphoric acid. [3]

One: ______________________________

Two: ______________________________

Three: ______________________________

Sodium hydroxide solution, $NaOH(aq)$, was used in a titration to determine the concentration of phosphoric acid.

(c) Other than it having too low a molar mass, state **two** reasons why the concentration of the sodium hydroxide solution cannot be reliably determined by weighing out an amount of solid sodium hydroxide and dissolving it in a known volume of distilled water. [2]

One: ______________________________

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Two: ______________________________

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The table below lists some acid-base indicators and the colour that each appears over a pH range.

Indicator	Colour		pH range
	Acid	Base	
Universal indicator	red	violet	1.0–14.0
Methyl orange	red	yellow	3.2–4.4
Bromocresol green	yellow	blue	3.8–5.4
Litmus	red	blue	4.5–8.3
Methyl red	yellow	red	4.8–6.0
Bromothymol blue	yellow	blue	6.0–7.6
Phenol red	yellow	red	6.8–8.4
Phenolphthalein	colourless	magenta	8.2–10.0

(d) Select the acid-base indicator from the table above that would be most suitable for the titration between phosphoric acid, $H_3PO_4(aq)$, and sodium hydroxide solution, $NaOH(aq)$. Justify your choice of indicator, including **one** relevant equation. [5]

9. [15 marks] *(2018:29)*

Wines and other alcoholic drinks can spoil when the alcohol (ethanol) they contain oxidises to acetic acid (ethanoic acid). An acidity regulator, monosodium citrate, is often added to drinks to prevent the formation of acetic acid. The monosodium citrate does this by acting as a buffer.

A citric acid/dihydrogen citrate ion buffer can be prepared from citric acid, $H_3C_6H_5O_7$ and monosodium citrate, $NaH_2C_6H_5O_7$.

(a) Write an equation for the buffer system ($H_3C_6H_5O_7/H_2C_6H_5O_7^-$) containing citric acid, $H_3C_6H_5O_7$ and monosodium citrate, $NaH_2C_6H_5O_7$. [2]

Buffers that contain equal concentrations of both components are most effective. This buffer solution is prepared by mixing 100.0 mL of citric acid solution with 100.0 mL of monosodium citrate solution. The citric acid solution, $H_3C_6H_5O_7(aq)$, has a concentration of 0.200 mol L^{-1}.

(b) Calculate the mass of sodium citrate, $NaH_2C_6H_5O_7$, that would need to be dissolved in 100.0 mL of distilled water to make the most effective buffer solution. [3]

(c) If a citric acid buffer was prepared to a pH of 3.5, what would be the concentration of the hydroxide ion at 25.0 °C? [3]

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(d) Explain why only a small change in pH is observed in this buffer solution when a small amount of sodium hydroxide solution is added, compared to adding a similar amount of sodium hydroxide solution to a system that is not a buffer solution. Your answer should refer to the buffer equilibrium in part (a). [4]

(e) Increasing the concentration of this buffer solution will increase its buffering capacity. Explain this statement. [3]

10. **[6 marks]** *(2018:31)*

Arsenic acid, $H_3AsO_4(aq)$, is a weak, triprotic acid that can be produced from the element directly through the reaction with water and ozone, $O_3(g)$. This reaction can be represented by the equation below.

$$2\,As(s) + 3\,H_2O(\ell) + 5\,O_3(g) \rightleftharpoons 2\,H_3AsO_4(aq) + 5\,O_2(g)$$

(a) Write the equilibrium constant expression for this reaction. [2]

(b) The arsenate ion, $HAsO_4^{2-}(aq)$, is amphoteric, meaning it can act as an acid and as a base.

(i) With the aid of equations, describe the amphoteric nature of $HAsO_4^{2-}$ in this aqueous solution. [3]

(ii) State why an aqueous solution containing $HAsO_4^{2-}$ is found to have a pH>7 at 25 °C. [1]

11. **[9 marks]** *(2019:26)*

Dilute hydrochloric acid, $HC\ell(aq)$, is added to three labelled test tubes.

(I) Excess copper metal, $Cu(s)$, is added to the first test tube.

(II) Excess copper(II) oxide, $CuO(s)$, is added to the second test tube.

(III) Excess copper(II) carbonate, $CuCO_3(s)$, is added to the third test tube.

(a) Describe the contents of the first and second test tubes once any reactions are complete. [4]

Test tube	Description
(I)	
(II)	

(b) Write the balanced equation, with appropriate state symbols, for the reaction that takes place between the copper(II) oxide and the hydrochloric acid. [3]

(c) If the labels of test tubes (II) and (III) became smudged, describe **all** the observations that could be used to distinguish between these test tubes once **any** reactions are complete. [2]

12. **[20 marks]** *(2019:36)*

The ideal pH of human blood is 7.4. If the pH of a person's blood varies too much from this value, a serious condition can develop. If the pH is too low, it is called acidosis; if the pH is too high, it is called alkalosis. Death may occur if the pH drops below 6.8 or rises above 7.8.

One buffer system for maintaining acid-base balance in blood is the carbonic acid-hydrogencarbonate buffer.

During exercise, the muscles need more oxygen to produce energy. They produce carbon dioxide, CO_2, and hydronium ions, H_3O^+, which move from the muscles to the blood.

The relevant equilibrium equations for the carbonic acid-hydrogencarbonate buffer system are shown as follows.

Equation 1 $H_3O^+(aq) + HCO_3^-(aq) \rightleftharpoons H_2CO_3(aq) + H_2O(\ell)$

(a) Identify the **two** conjugate acid-base pairs on Equation 1 above, indicating clearly which is the acid and which is the base in each pairing. [2]

(b) Write the equilibrium constant expression for Equation 1. [2]

Carbonic acid further reacts to form water and carbon dioxide as shown in Equation 2.

Equation 2 $H_2CO_3(aq) \rightleftharpoons H_2O(\ell) + CO_2(aq)$

(c) Combine Equations 1 and 2, to create an overall equation that shows the relationship between $HCO_3^-(aq)$ and $CO_2(aq)$. [2]

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(d) Identify the effect on the blood's pH when each of the following components are removed: carbon dioxide and hydrogencarbonate ions. [2]

Component removed	Effect on pH (circle your answer)		
carbon dioxide	increase	decrease	no effect
hydrogencarbonate ions	increase	decrease	no effect

The buffering capacity of the carbonic acid-hydrogencarbonate is greatest when the pH is between 5.1 and 7.1.

(e) State **two** conditions in terms of concentration that are necessary for this buffering capacity to be optimal. [2]

One: ____

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Two: ____

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When the pH of the blood is too high, the kidneys can remove hydrogencarbonate ions, HCO_3^-, from the blood.

(f) Use Le Châtelier's Principle to demonstrate that the kidneys' action can help to prevent excessively high blood pH. [3]

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When inhaling, oxygen is taken into the lungs and transferred to the blood; when exhaling, carbon dioxide is expelled.

During hyperventilation (very rapid and deep breathing) more carbon dioxide is being expelled from the body than it can produce. This upsets the oxygen/carbon dioxide balance and can cause dizziness and fainting. Hyperventilating results in lowering the carbon dioxide concentration in the blood, which can affect the pH of the blood.

The equation shown below illustrates the formation of hydronium ions within the blood system.

$$2\,H_2O(\ell) + CO_2(aq) \rightleftharpoons H_3O^+(aq) + HCO_3^-(aq)$$

A first-aid treatment for hyperventilation is the 'paper-bag treatment' whereby the patient breathes into a paper bag and so breathes back in the expelled breath, which contains a higher concentration of carbon dioxide.

(g) State the effect of the 'paper-bag treatment' on the pH of the blood and explain why it is an effective treatment for hyperventilation. [3]

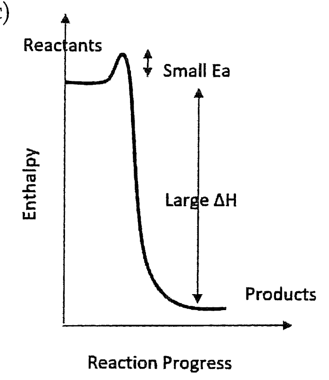
Another contributor to a potential imbalance of blood pH is the formation of lactic acid. The chemical name for lactic acid is 2-hydroxypropanoic acid, $C_3H_6O_3$.

(h) Draw the structural formula for lactic acid with **all** its functional groups circled and labelled. [4]

When you complete an answer to a question, read the question again and then you see you need to state that the pH decreases and then write this equation.

$$OC\ell^-(aq) + H_3O^+(aq) \rightleftharpoons HOC\ell(aq) + H_2O(\ell)$$

c)



Chapter 8: Acids and Bases

1.(2015:29) a) $c(H^+) = n/v = 8.50 \times 10^{-3}/ 0.025 = 0.340$ mol L^{-1}

$pH = -\log[H^+] = -\log(0.340) = 0.469$

b) $n(OH^-) = cv = 0.3 \times 0.02 = 0.006$ mol

OH^- and H^+ react in 1:1 ratio

$n(H^+)$ in excess = $0.0085 - 0.006 = 2.5 \times 10^{-3}$ mol

$c(H^+) = 2.5 \times 10^{-3}/0.045 = 0.0556$ mol L^{-1}

$pH = -\log [H^+] = -\log(0.0556) = 1.26$

2.(2015:40) b) The same pH means the same $[H^+]$. HF is a weak acid and does not ionise to the same extent as $HC\ell$. So a greater concentration of HF is needed to give the required $[H^+]$

c) Fluoride hydrolyses resulting in the formation of hydroxide ions producing a solution with a pH >7

$F^-(aq) + H_2O(\ell) \rightleftharpoons HF(aq) + OH^-(aq)$

The chloride ion is the weak conjugate base of a strong acid ($HC\ell$) and is a neutral ion

d) i) Hydrolysis of propanoic acid:

$CH_3CH_2COOH + H_2O \rightleftharpoons CH_3CH_2COO^-(aq) + H_3O^+(aq)$

When hydrogen ions are added to the buffer, the equilibrium will (by LCP) shift left to use up the added H^+ ions. So the overall change to $[H^+]$ is minimised and so the pH change is insignificant.

ii) Equal concentrations of acid / conjugate base and high concentrations of acid and conjugate base.

3.(2016:31) a)

Acidic	Neutral	Basic
$NH_4C\ell$	$KC\ell$, $NaNO_3$	KCN, $Mg_3(PO_4)_2$

b) This question has two key parts you must address. It asks you to use Brønsted-Lowry theory (so $NH_3 + H^+ \rightarrow NH_4^+$ is not adequate) and to explain why the pH of ammonia solution is greater than 7.0 (you must explain the significance of these numbers and not just say it's basic).

Example:

$NH_3(g) + H_2O(\ell) \rightleftharpoons NH_4^+(aq) + OH^-(aq)$ – note the states are included and the equilibrium arrow is used.

Brønsted-Lowry discussion: Ammonia acts as a weak base accepting a proton from water, acting as an acid, to form the ammonium ion and hydroxide ions.

pH discussion: In water the concentration of H^+ and OH^- are both 10^{-7} mol L^{-1} This reaction means the concentration of OH^- will increase above the H^+ so the pH will be above 7.

4.(2016:32) a) $H_2CO_3(aq) + H_2O(\ell) \rightleftharpoons H_3O^+(aq) + HCO_3^-(aq)$ – note the equilibrium arrow.

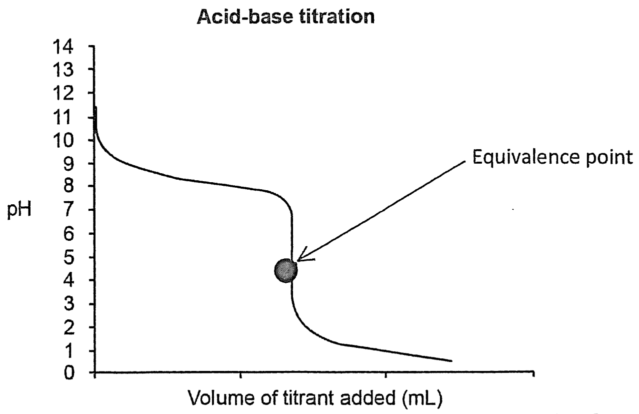
b) This question tests your understanding of buffers. Most students know a buffer is a mixture of a weak acid (or base) and its conjugate in high concentration eg 1 mol L^{-1}.

In this question the H_2CO_3/HCO_3^- meets this requirement. Here $HC\ell$ forms H_3O^+ in water

$HC\ell(aq) + H_2O(\ell) \rightleftharpoons H_3O^+(aq) + OH^-(aq)$

The HCO_3^- accepts protons from H_3O^+ to form H_2CO_3 removing the newly added H_3O^+ions and so there is little change in the pH because the BUFFER CAPACITY has not been reached. But in the case of water the species are at 10^{-7} mol L^{-1} concentration. Thus their BUFFER CAPACITY is small. Hence the very poor buffer is easily overwhelmed by the addition of the small quantity of $HC\ell$ which increases the concentration of H_3O^+ ions, lowering the pH.

5.(2016:34) a)



b) The syllabus does not mention any indicators you need to know specifically. This question requires you to know how to pick an indicator for a titration.

We have a weak base/strong acid titration which has an equivalence point at about pH 4.5. So we need an indicator with an end point of 4.5.

Bromocresol Green because ...

- Its colour change is at a suitable end point pH to show the equivalence point has been reached
- That a change in colour of the indicator (end point) indicates that the equivalence point has been reached
- Other indicators like Phenolphthalein, Alizarin Yellow R change too early while others like Methyl Yellow, Bromocresol Blue change too late and miss the equivalence point)
- Wrong indicator use will not be corrected by repeating the experiment therefore wrong indicator use is a systematic error

6.(2017:31) a) $2\ H_2O(\ell) \rightleftharpoons H_3O^+(aq) + OH^-(aq)$

b) $K = [H_3O^+][OH^-]$

c) It tells us that there are very few ions produced. The ratio of reactant to product is huge. The constant says nothing about the rate of reaction.

d) At 50°C, $K = 5.48 \times 10^{-14}$

Since $K = [H^+] \times [OH^-]$ and they are equal then $[H^+] = \sqrt{5.48 \times 10^{-14}} = 2.34 \times 10^{-7}$

$pH = -\log 2.34 \times 10^{-7} = 6.63$

e) Water is neutral because $[H^+] = [OH^-]$ at all temperatures

7.(2018:26) a) As carbonic acid is weak we write it as $H_2CO_3(aq)$ not $H^+(aq)$ and $CO_3^{2-}(aq)$.

$Cu(OH)_2(s) + H_2CO_3(aq) \rightarrow CuCO_3(s) + 2\ H_2O(\ell)$

b) A blue solid is mixed with a colourless, clear liquid. A green solid precipitate appears. $CuCO_3(s)$ is green).

c) The colourless solution would turn blue and there would be no solid remaining.

d) Select two from safety glasses, gloves, lab coats, closed shoes, tie back hair.

8.(2018:27) a) $H_2PO_4^-(aq) + H_2O(\ell) \rightleftharpoons HPO_4^{2-}(aq) + H_3O^+(aq)$.

b) There are a large number of possible answers to this question. You might list three ions H^+, H_3O^+, Mg^{2+}, OH^-, HCO_3^-, CO_3^{2-}, PO_4^{3-}, HPO_4^{2-}, $H_2PO_4^-$ or you might list insoluble solids such as, $MgCO_3$, $Mg_3(PO_4)_2$ or soluble substances such as H_2CO_3, or CO_2

c) Hydroscopic (absorbs water from the air), deliquescent (absorbs water from the air and forms a solution) or reacts with carbon dioxide in the air to form sodium carbonate.

d) The best answer is phenolphthalein. In the neutralisation reaction sodium phosphate (*aq*) would be formed after the weak phosphoric acid is neutralised. Because it is weak it won't lose its final proton until a high pH – maybe 9? Phosphate is the salt of the weak acid phosphoric acid. Therefore, the phosphate ion undergoes hydrolysis like this:

$$PO_4^{3-}(aq) + H_2O(\ell) \rightleftharpoons HPO_4^{2-}(aq) + OH^-(aq)$$

Phenolphthalein has an end point appropriate for this equivalence point.

9.(2018:29) a) $H_3C_6H_5O_7(aq) + H_2O(\ell) \rightleftharpoons H_2C_6H_5O_7^-(aq) + H_3O^+(aq)$.

b) If the citric acid is 0.200 mol L^{-1} then the monosodium citrate should also be 0.200 mol L^{-1}

$n\ (H_2C_6H_5O_7^-) = c.V = 0.200 \times 0.100 = 0.0200$ mol

$m\ (NaH_2C_6H_5O_7) = n.M = 0.0200 \times 214.1054 = 4.282108 = 4.28$ g (3 sf)

c) The simplest way to make this calculation is to remember pH + pOH = 14 (at 25 °C)

$\therefore$ pOH = 14 – 3.5 = 10.5

$\therefore [OH^-] = 10^{-10.5} = 3.16 \times 10^{-11}$ mol L^{-1}

d) This question has two parts. To clarify your thoughts, use headings.

Buffer

The small amount of sodium hydroxide reacts with the citric acid;

$H_3C_6H_5O_7(aq) + OH^-(aq) \rightarrow H_2C_6H_5O_7^-(aq) + H_2O(\ell)$

and is removed by the acidic component of the buffer producing citrate ions and uses up some citric acid.

The decreased concentration of citric acid means the concentration of H_3O^+ ions will decrease and the pH will rise a small amount.

Non-buffer

When OH^- is added to a system that is not a buffer solution there is no acid to neutralise the base and the $[OH^-]$ rises as does the pH

e) Increasing the concentration of the buffer involves increasing the concentration of both the weak acid and the conjugate. This means the system is able to react with more acid or base before a significant change to pH occurs. This is the definition of buffer capacity.

10.(2018:31) a) $K = \dfrac{[H_3AsO_4]^2[O_2]^5}{[O_3]^5}$

b) i) The arsenate ion, $HAsO_4^{2}(aq)$ can gain or lose a proton. This is called amphoteric:

$HAsO_4^{2-} + H_2O \rightleftharpoons H_2AsO_4^- + OH^-$

$HAsO_4^{2-} + H_2O \rightleftharpoons AsO_4^{3-} + H_3O^+$.

ii) If the solution has a pH greater than 7 it means that the first equation, above, has a greater K value i.e. the first equation produces more product than the second equation.

11.(2019:26) a) This question is all about reading the question closely. The word 'excess' is very important – in chemistry questions every word is important.

i) The reaction of acid and copper is not spontaneous – there is no reaction. You must, however, describe all observations. A salmon pink (you must use the colours in the data book) solid in a 'colourless clear liquid' or 'colourless solution'.

ii) Copper oxide is black and in excess. So there is black powder present but some has reacted with the hydrochloric acid to make a copper chloride solution which will be blue. Interesting fact that copper chloride is a green solid but forms a blue solution. In general, the rule is – solutions are the colour of the ion and this example bears that out. In this case green solution would probably be acceptable.

b) Remember always to write ionic equations.

$$CuO(s) + 2\,H^+(aq) \rightarrow Cu^{2+}(aq) + H_2O(\ell)$$

c) This question refers again to the use of the word 'excess'. Tube 1 has excess solid black copper oxide and tube 3 has excess solid green copper carbonate.

12.(2019:36) a)

$$\underset{\text{Acid 1}}{H_3O^+(aq)} + \underset{\text{Base 2}}{HCO_3^-(aq)} \rightleftharpoons \underset{\text{Acid 2}}{H_2CO_3(aq)} + \underset{\text{Base 1}}{H_2O(\ell)}$$

b) $K = \dfrac{[H_2CO_3]}{[HCO_3^-][H_3O^+]}$

c) We always write ionic equations

$H_3O^+(aq) + HCO_3^-(aq) \rightleftharpoons 2\,H_2O(\ell) + CO_2(aq)$

d)

Component removed	Effect on pH (circle your answer)		
carbon dioxide	increase	decrease	no effect
hydrogencarbonate ions	increase	decrease	no effect

e) That the conjugate pair that make up the buffer (hydrogencarbonate ion and carbonic acid) are of equal and high concentration.

f) Using Le Châtelier's Principle means we do not discuss forward and backward rates and collisions but instead refer to overcoming a constraint to re-establish equilibrium.

Given that this equilibrium existed.

$H_3O^+(aq) + HCO_3^-(aq) \rightleftharpoons 2\,H_2O(\ell) + CO_2(aq)$

When the kidneys applied a constraint by removing hydrogencarbonate ions the equation moved to the left to re-establish equilibrium. In doing so more hydronium ions are formed. So the $[H_3O^+]$ increases and the pH decreases.

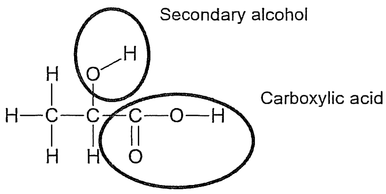
g) This question uses the term 'explain'. The syllabus specifies that 'explaining' can only be done with collision theory. Thus LCP is not acceptable here.

Breathing into a paper bag and re-inhaling the same air increases the $[CO_2]$ in the blood.

$H_3O^+(aq) + HCO_3^-(aq) \rightleftharpoons 2\,H_2O(\ell) + CO_2(aq)$

Therefore, the frequency of collisions of the products will increase and there will be a higher frequency of successful collisions. The reverse rate of reaction will increase. The forward rate initially remains the same so the position of equilibrium moves to the left. This increases $[H_3O^+]$ and the pH of the blood is lowered overcoming the pH rising effect of hyperventilation.

h)



Chapter 9: Oxidation-Reduction

1.(2014:31) a) The role of the standard hydrogen half cell in determining the table of the Standard Reduction Potentials.

Standard Reduction potentials (E°) are measured with solutions of 1.0 mol L^{-1} concentrations, (with gases at a pressure of one atmosphere) and at a temperature of 25°C. They are a measure of the relative capacity of the more oxidisable form of the couple to gain electrons, and are called reduction potentials. The hydrogen half cell is assigned a E° value of exactly zero volt because this is used as a reference reaction. Standard Reduction Potentials for all other reduction reactions are obtained by measuring the total cell voltage or emf when attached to a standard hydrogen half cell. Thus H – half cell, acts as a standard reference for all other cells.

Example: If a zinc half cell is connected to a hydrogen half cell, (under standard conditions) the voltmeter (which measures the emf or cell potential) will read 0.76 V. The cell reaction is

$Zn(s) + 2\,H^+(aq) \rightarrow Zn^{2+}(aq) + H_2(g)$

Because zinc is oxidised and donates electrons to hydrogen through the external circuit, the oxidation half reaction is $Zn(s) \rightarrow Zn^{2+}(aq) + 2\,e^-$. This has an *oxidation potential* of 0.76 V.

Therefore, the reduction half reaction, $Zn^{2+} + 2\,e^- \rightarrow Zn(s)$ has a standard reduction potential of –0.76 V.

In the Standard Reduction Potential table, the half reactions are arranged in the order of decreasing reduction potentials.

b) Limitations of the Reduction Potential table:

- The half-reactions given in the table apply to aqueous solutions only.
- The E° values are given for concentrations of 1.0 mol L^{-1}.
- The electrode reactions depend on the nature of the electrode surface.
- The table does not provide any information on the rate of reactions.
- Only applies to 25°C and 101 kPa.
- Experimentation is the only way to make sure that a reaction will occur.

2.(2014:32) Step 1: Reduction of nitrogen gas to ammonia.

$N_2(g) + 8\,H^+(aq) + 6\,e^- \rightarrow 2\,NH_4^+(aq)$

Step 2: Oxidation of ammonium ions to nitrite ions.

$NH_4^+(aq) + 2\,H_2O(\ell) \rightarrow NO_2^-(aq) + 8\,H^+(aq) + 6\,e^-$

Step 3: Oxidation of nitrite ions to nitrate ions.

$NO_2^-(aq) + H_2O(\ell) \rightarrow NO_3^-(aq) + 2\,H^+(aq) + 2\,e^-$

3.(2014:36) a) The independent variable (the variable that is altered at the discretion of the investigator) is the concentrations of the solutions.

b) The dependent variable (the variable that is affected as a direct result) is the electrical potential or voltage of the cell.

c) All the other variables which can have an impact on the electrical potential such as the volume of the solution, the temperature of the solution and the surface area of the electrodes – need to be kept constant. This is to enhance the reliability of the findings in this investigation.

d) Explanation for the increase in electrical potential due to changes in the concentrations of Cu^{2+} ions and Zn^{2+} ions. In general, in any electrochemical cell, if the concentration of the reactants increases relative to the concentration of the products, the cell reaction becomes more spontaneous and the cell emf increases. An increase in the concentration of the reactant ions (Cu^{2+}) increases the rate of the forward reaction. If the concentration of the products (Zn^{2+}) increases relative to the reactants, the cell emf decreases. An increase in the concentration of the product ions decreases the rate of the forward reaction and increases the rate of the reverse reaction.

e) As the cell operates, the concentration of the reactant ions decreases. This decreases the rate of the forward reaction and the cell emf decreases.

f) Two ways to improve the investigation:

- Since different cells require different electrolytes, it is important to test a number of different cells with different electrolytes to verify if the relationship is general to all cells.
- For the same cell, evidence from one set of data will not be sufficient to make the relationship reliable. Results from at least a few other trials would be necessary to verify the relationship.

4.(2014 S2:40a,c) a) i) Draw arrow downward towards the copper cathode.

ii) Arrow drawn pointing to the right towards the copper cathode.

b) Electrolysis requires the movement of charge around the complete circuit, (electrons in the wiring and electrodes) and ions in the electrolyte. In the solid state, the ions within copper(II) sulfate are not free to move so no current can flow. When dissolved in water the copper(II) ions and sulfate ions are now free to move through the electrolyte and so complete the circuit.

5.(2015:31) a) $2\,Ag^+_{(aq)} + CO_3^{2-}{}_{(aq)} \rightarrow Ag_2CO_{3(s)}$